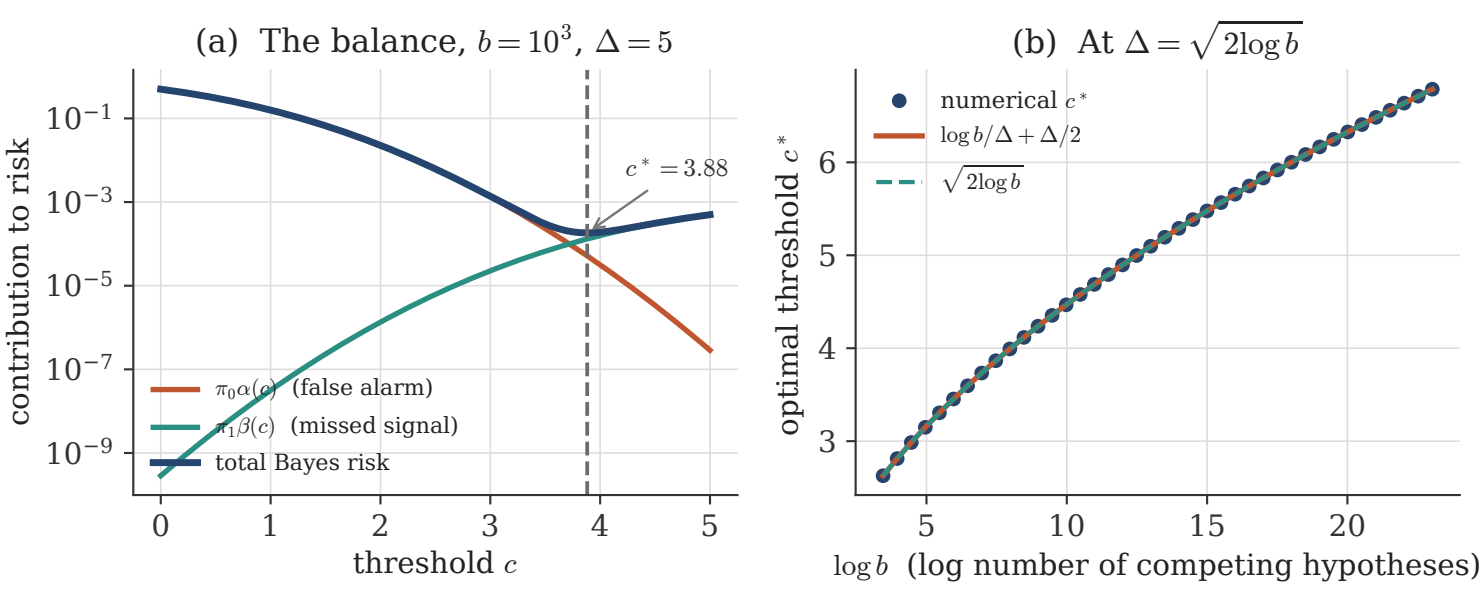
(a) The balance, $b = 10^3$, $\Delta = 5$
contribution to risk
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
$c^* = 3.88$
$\pi_0\alpha(c)$ (false alarm)
$\pi_1\beta(c)$ (missed signal)
total Bayes risk
0
1
2
3
4
5
threshold c
(b) At $\Delta = \sqrt{2\log b}$
optimal threshold c^*
numerical c^*
$\log b/\Delta + \Delta/2$
$\sqrt{2\log b}$
3
4
5
6
5
10
15
20
$\log b$ (log number of competing hypotheses)